

COMPUTER SCIENCE PROJECT

MICROSOFT EXCEL

Functions in Excel: Text & Numerical Functions

SUBMITTED BY

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CLASS

7

ROLL NO.

01

SUBJECT

Computer Science

SCHOOL

JTS English Medium School

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01

Introduction

Microsoft Excel is one of the most popular and powerful spreadsheet software applications developed by Microsoft. It is widely used for storing, organising, calculating, and analysing data in a tabular format, arranged in the form of rows and columns intersecting each other, forming boxes called ‘cells’. Because of its flexibility, simplicity, easy-to-use functions, and ability to integrate AI, it is used almost everywhere to manage and analyse data. It can also perform advanced mathematical operations in seconds, along with creating graphs and charts.

What Excel does

Store

Keeps data in a tabular format.

Organise

Arranges data in rows and columns.

Calculate

Performs advanced maths in seconds.

Analyse

Creates graphs and charts from data.

Rows, columns and cells

	A	B	C	D	E
1					
2					
3			C3		
4					
5					

Row (runs across)

Column (runs down)

Cell (row + column)

02

Functions in Excel

Functions are pre-defined formulas in Excel that help us perform simple as well as complex calculations. They are capable of accepting arguments and returning values. The arguments are written inside parentheses, which may contain text or numerical input values. A function begins with a compulsory equal-to (=) sign, followed by the function name and arguments inside parentheses. E.g. “=SUM(6,9971)” gives an output of “9977”.

ANATOMY OF A FUNCTION



=

Equal-to sign (compulsory, always first)



SUM

Function name



(6, 9971)

Arguments, written inside parentheses

Output: 9977

GENERAL SYNTAX

=FUNCTION(arguments)

Functions can perform complex arithmetic, trigonometric, and logical calculations, as well as work with text entries.

Text Functions

10 functions that work with text entries (page 5)

Numerical Functions

10 functions for arithmetic calculations (page 6)

03

Text Functions in Excel

Function	Purpose	Example	Output
LEFT (text, [num_chars])	Extracts characters from the left side of text.	=LEFT("COMPUTER",4)	COMP
RIGHT (text, [num_chars])	Extracts characters from the right side of text.	=RIGHT("COMPUTER",4)	UTER
MID (text, start_num, num_chars)	Extracts characters from the middle of text.	=MID("COMPUTER",3,4)	MPUT
LEN (text)	Counts the number of characters in text.	=LEN("Excel")	5
UPPER (text)	Converts text into uppercase.	=UPPER("excel")	EXCEL
LOWER (text)	Converts text into lowercase.	=LOWER("EXCEL")	excel
PROPER (text)	Capitalises the first letter of each word.	=PROPER("microsoft excel")	Microsoft Excel
TRIM (text)	Removes extra spaces from text.	=TRIM(" Hello World ")	Hello World
CONCAT (text1, [text2], ...)	Joins two or more text strings.	=CONCAT("Hello"," World")	Hello World
TEXT (value, format_text)	Converts a number into formatted text.	=TEXT(1234.5,"0.00")	1234.50

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Numerical Functions in Excel

Function	Purpose	Example	Output
SUM (number1, [number2], ...)	Adds numbers together.	=SUM(10, 20, 30)	60
AVERAGE (number1, [number2], ...)	Calculates the average of numbers.	=AVERAGE(10, 20, 30)	20
MAX (number1, [number2], ...)	Finds the largest value.	=MAX(10, 25, 15)	25
MIN (number1, [number2], ...)	Finds the smallest value.	=MIN(10, 25, 15)	10
COUNT (value1, [value2], ...)	Counts cells containing numbers.	=COUNT(10, 20, 30)	3
PRODUCT (number1, [number2], ...)	Multiplies numbers together.	=PRODUCT(5, 4)	20
ROUND (number, num_digits)	Rounds a number to specified digits.	=ROUND(12.567, 2)	12.57
ABS (number)	Returns the absolute value of a number.	=ABS(-25)	25
SQRT (number)	Calculates the square root of a number.	=SQRT(64)	8
POWER (number, power)	Raises a number to a specified power.	=POWER(5, 2)	25